

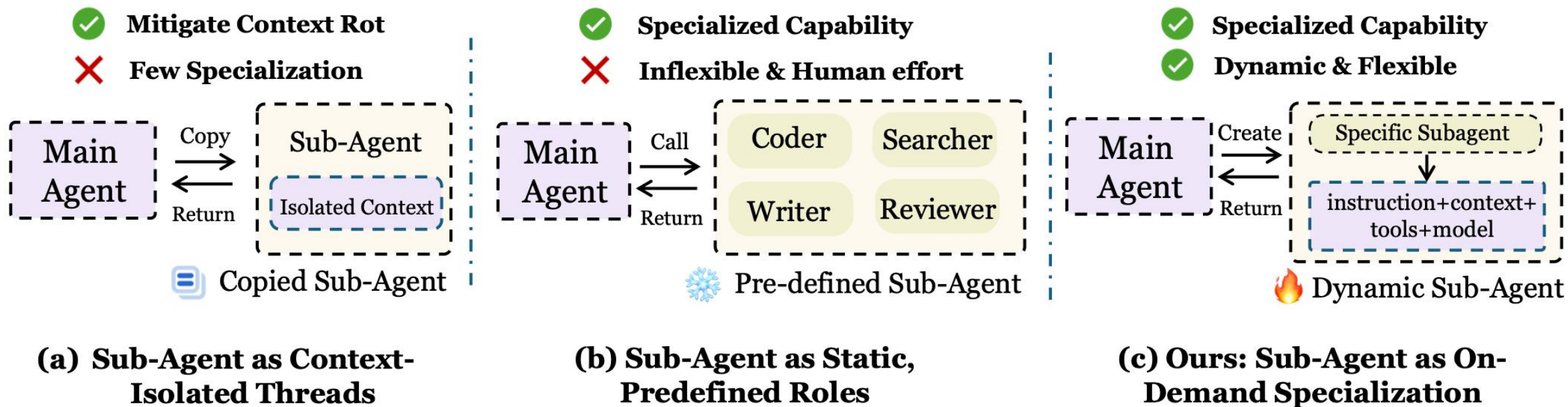
AORCHESTRA: Automating Sub-Agent Creation for Agentic Orchestration

ICML 2026

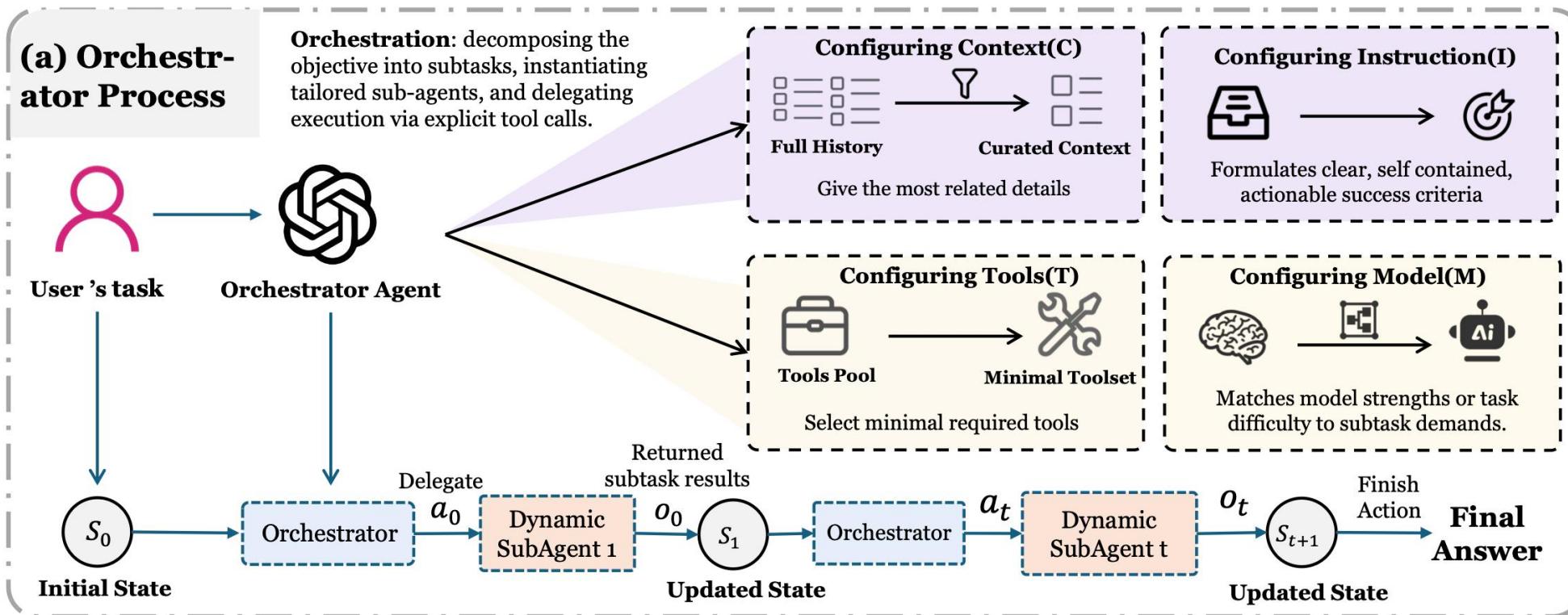
研究背景:

- **多智能体范式有益**: 多智能体系统能够应对日益复杂的场景, 表现超越单一智能体
- **多智能体的协调开销**: 信息的过度共享导致噪声/关键信息遗漏, 使得难以稳健地长期运行
- ***Subagent-as-tools*范式的出现**: 为解决多智能体的上述困境。主智能体 (*Orchestrator*) 通过显式的工具调用将任务委托给子智能体 (*subagent*)

对比不同的sub-agent-as-tools方法:

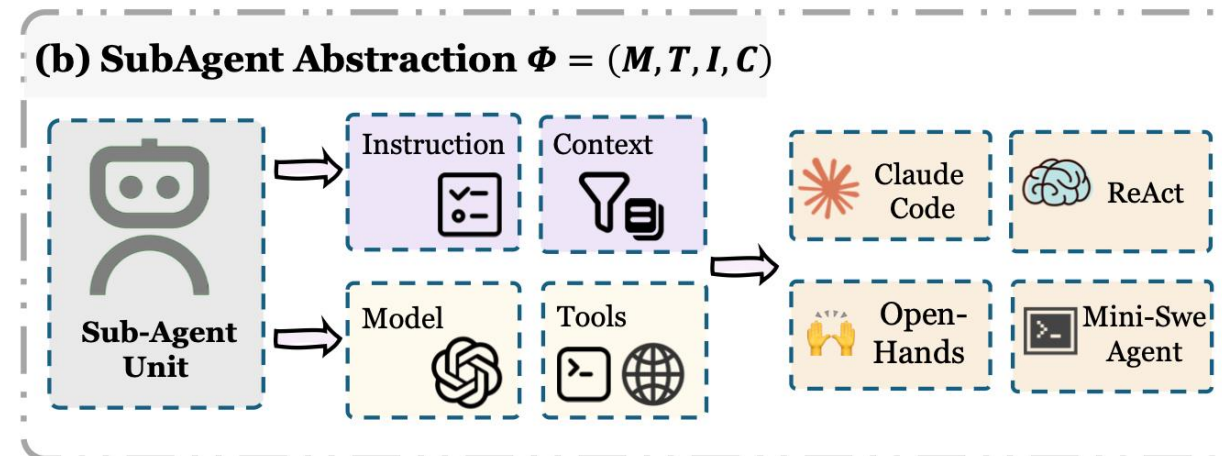
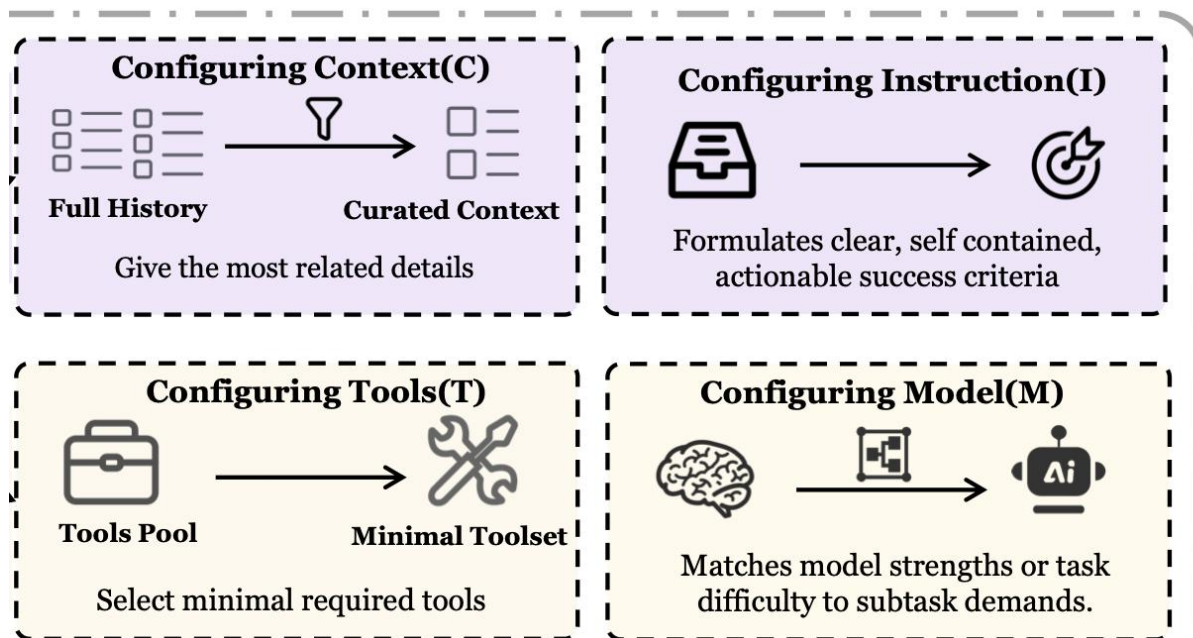


- (a) 子智能体作为上下文隔离的单独进程：避免上下文的污染，但是缺乏对需求的定制化
- (b) 子智能体作为预定义的静态角色：无法覆盖开放环境中动态涌现的各种子任务，需要较大的人工工程量
- (c) 按需定制子智能体：兼顾定制化的需求以及动态灵活性



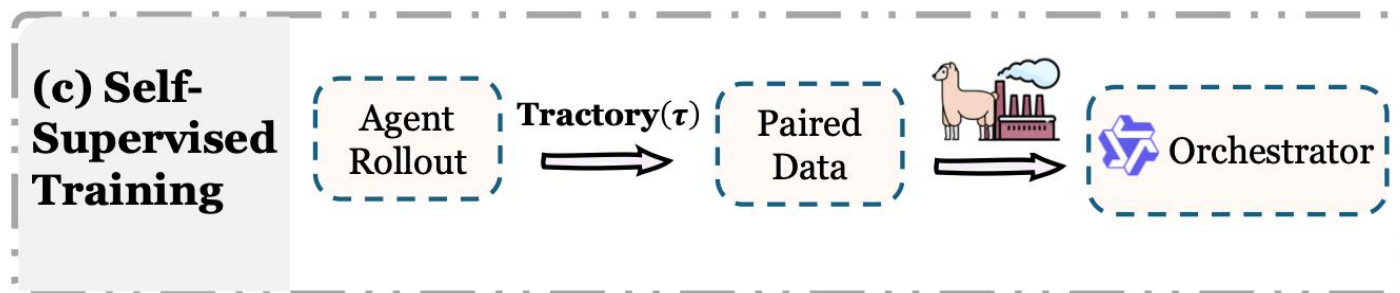
- **Subagent-as-tools循环**: 将此经典的 *think&tool-call* 循环, 转变为 *think&subagent* 循环
- **主智能体 Orchestrator 的动作空间**: 仅包含 “派发” 和 “终止” 这两种动作

$$\mathcal{A}_{orch} \supseteq \mathcal{A}_{env} \cup \{\text{Delegate}(\cdot), \text{Finish}(y)\}.$$

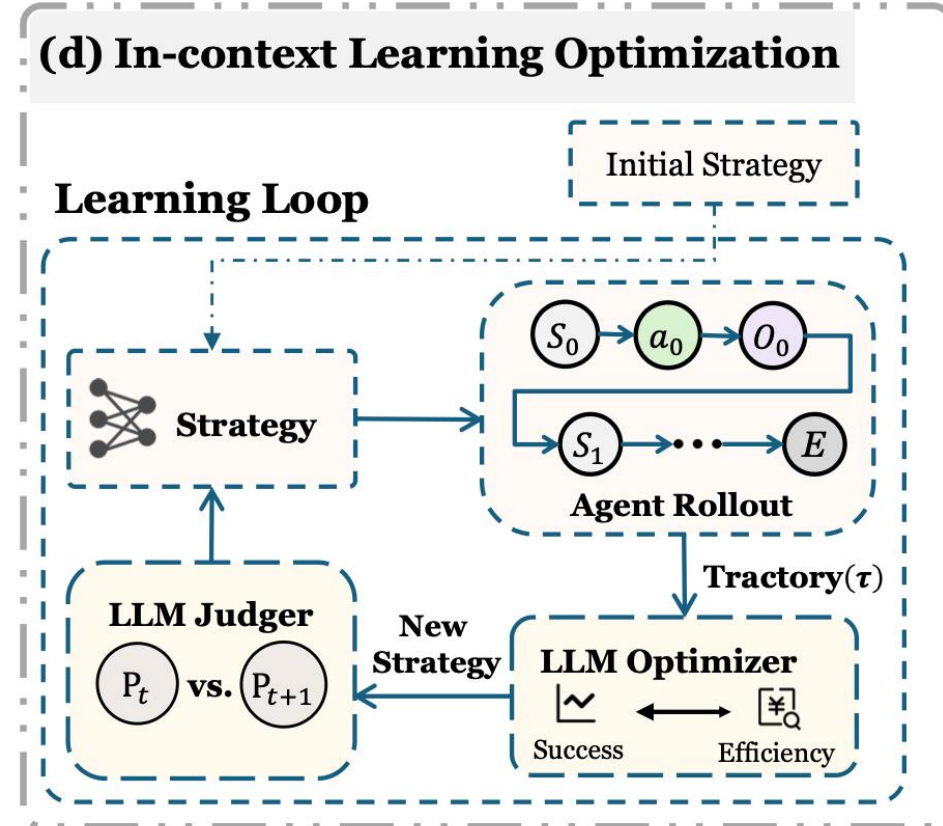


- **任意智能体被抽象为四元组**: 上下文, 指令, 可用工具, 模型
- **Orchestrator在派发子智能体时**: 给出指令, 裁剪上下文, 选择合适的工具集, 选择模型
- **子智能体仍可选用多样的范式/Harness**: ReAct, Claude Code, Open-Hands

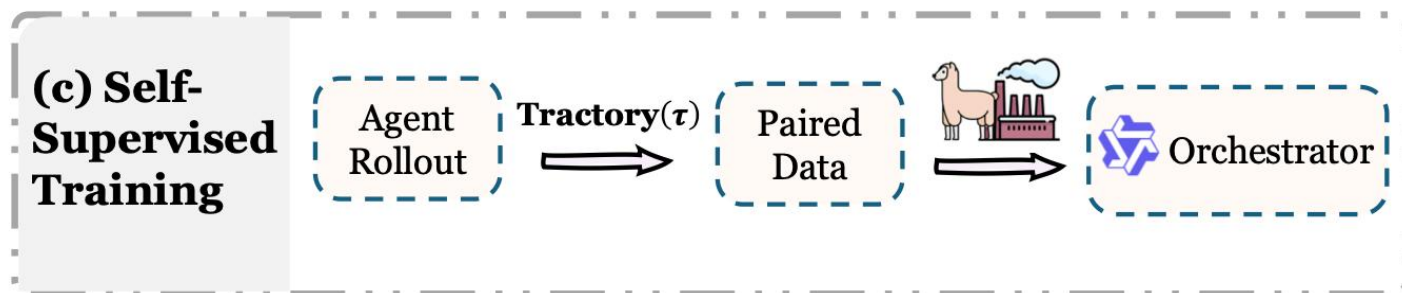
Methods



- **Orchestrator的策略优化**: 对Task Orchestration进行优化, 对Model Selection进行优化
- **对Task Orchestration**: 给出更好的指令、上下文, 工具子集。采用SFT方法。
- **对Model Selection**: 选择更合适的Model以平衡 performance-cost。采用Prompt ICL Optimization方法。

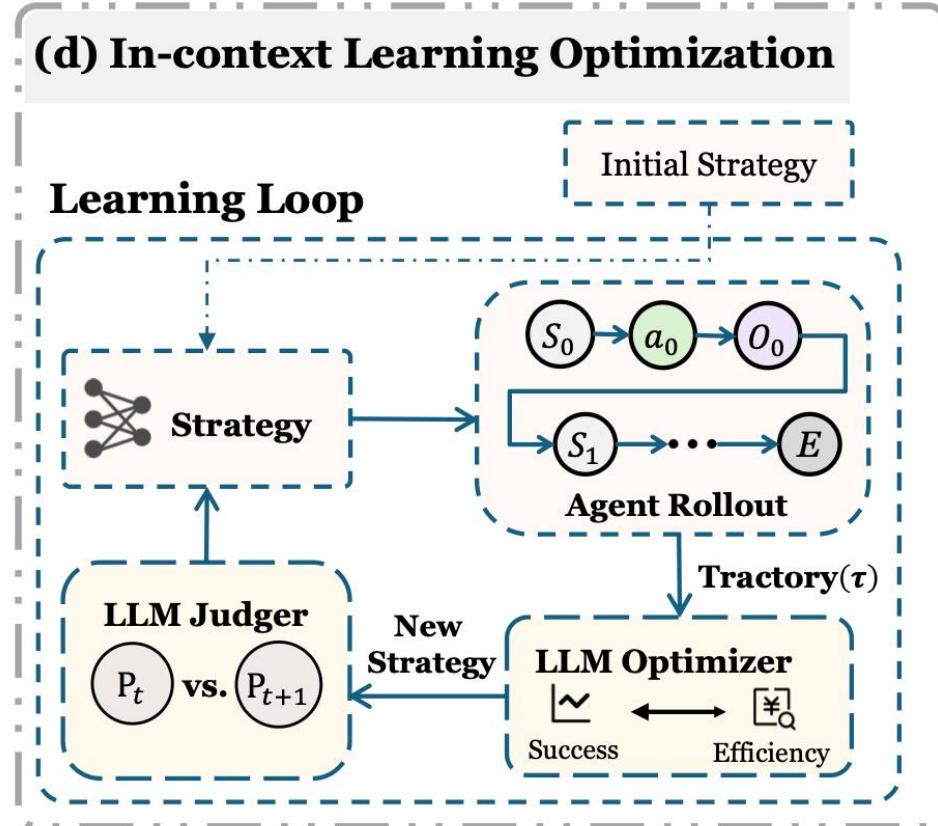


Methods



- **SFT方法**: 在TaskCraft数据集上, 使用Gemini-3-flash获取2000条轨迹, 全参数训练两轮Qwen-3-8B non thinking
- **Prompt ICL Optimization方法**: 通过另一LLM (Claude-4.5-sonnet) 作为优化器, 根据轨迹、任务指标、Token开销优化Orchestrator的 prompt. 优化5轮, 每一轮使用6条数据

$$I_{k+1}^{\text{main}} = \text{OPTIMIZE}(I_k^{\text{main}}, \tau_k, \text{Perf}(\tau_k), \text{Cost}(\tau_k)),$$



- **Baselines:** ReAct, OpenHands, mini-SWE-agent, Claude Code
- **Benchmarks:** Terminal Bench 2.0, SWE Bench Verified, GAIA
- **Implementation:** 对于任何baselines, 设置max steps=500保证公平性。对于本文方法, SubAgent的max step=50, 外层Orchestrator的max attempt=10
- **AOrchestrator:** 使用Gemini-3-flash作为Orchestrator, Subagent的核心LLM仅有一种模型, Subagent在第1、3个Bench上使用ReAct, 在第2个bench上使用SWEBenchSubAgent

Experiments

Methods	Model Setup	GAIA		Terminal-Bench 2.0		SWE-Bench-Verified		
		Pass@1	Pass@3	Pass@1	Pass@3	Pass@1	Pass@3	Avg. Pass@1
ReAct	Gemini-3-Flash	49.09	66.06	28.57	47.14	64.00	82.00	47.22
	DeepSeek-V3.2	46.70	71.51	20.00	32.86	48.00	87.00	38.23
	Claude-4.5-haiku	47.88	62.42	20.00	37.14	63.00	87.00	43.62
OpenHands	Gemini-3-Flash	66.06	72.73	31.43	51.43	48.00	66.00	48.49
	DeepSeek-V3.2	63.64	72.12	21.43	35.71	60.00	75.00	48.35
	Claude-4.5-haiku	54.55	61.21	12.85	25.71	68.00	83.00	45.13
Mini-SWE	Gemini-3-Flash	58.18	68.48	34.29	50.00	56.00	85.00	49.49
	DeepSeek-V3.2	50.30	63.63	30.00	48.57	84.00	89.00	54.76
	Claude-4.5-haiku	40.61	60.00	24.29	28.57	44.00	83.00	36.30
Claude Code	Gemini-3-Flash	–	–	32.86	48.57	22.00	42.00	27.43
	Claude-4.5-haiku	–	–	34.29	45.71	25.00	41.00	29.65
AORCHESTRA	Gemini-3-Flash	80.00	86.06	52.86	57.14	82.00	86.00	71.62
	DeepSeek-V3.2	67.87	80.00	31.43	42.86	76.00	82.00	58.43
	Claude-4.5-haiku	60.61	73.90	35.71	45.71	70.00	84.00	55.44

- 对于Aorchestra, Model Step指的是Subagent使用的Model
- 在相同core LLM下产生较为一致的提升

Table 2. Context-control ablation for sub-agent invocation. We isolate the effect of context inheritance by only changing the Context field passed to sub-agents, while keeping the sub-agent model, tools, and system prompt identical across settings.

Setting	Level 1	Level 2	Level 3	Avg.
No-Context	89.47	81.48	75.00	86.00
Full-Context	94.74	77.78	75.00	84.00
Ours	100.00	88.89	75.00	96.00

- 研究Orchestrator派发给子智能体的上下文的影响
- Non-Context: 子智能体仅获得Instruction
- Full-Context: 子智能体获得Instruction, 以及Orchestrator的全部上下文
- 在GAIA基准上进行消融, Level-x指任务难度, 越高难度越大 (有点小问题)

Table 3. Main results. ReAct is evaluated with a single specified LM per run, while other systems may use either a single LM or a mixed-LM pool. ICL denotes context learning

System	LM	Acc.	Avg. Cost
ReAct	Claude-4.5-sonnet	53.93	0.190
	Claude-4.5-haiku	47.88	0.066
	Gemini-3-Flash	49.09	0.070
	GPT-5-mini	54.55	0.052
	Deepseek-v3.2	46.70	0.027
Ours	Gemini-3-Flash	80.00	0.79
	Claude-4.5-sonnet	71.52	0.91
	GPT-5-mini	67.27	0.28
	Deepseek-v3.2	67.87	0.14
Ours (Gemini-3-Flash)	Mixed	72.12	0.70
Ours (ICL)	Mixed	75.15	0.25
Ours (Qwen3-8B)	Gemini-3-Flash	56.97	0.36
Ours (SFT)	Gemini-3-Flash	68.48	0.68

- 任务性能多大程度依赖于
Orchestrator: 80.00-->56.97,
较弱的Orchestrator显著削弱性能
- SFT训练后的Orchestrator可缓解:
56.97-->68.48
- Case Study发现, SFT主要提升两方面表现:
 - 1. 在Subagent没有得到明确答案时, Orchestrator学会从轨迹中自己寻找答案
 - 2. Orchestrator不再过度依赖首次Subagent获取的答案

Experiments

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- SubAgent使用不同的LLM会获得不同的Accuracy-Cost曲线
- Mixed指SubAgent可以从LLM Pool中选择一种LLM，而不是只能使用唯一LLM
- **ICL使得Orchestrator能够降本增效：** 72.12-->75.15, 0.70→0.25
- Mixed却在绝对性能上弱于单一设置： 75.15 vs 80.00

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- 通过case分析，ICL对Orchestrator的行为影响：
 - 1. ICL优化过的Orchestrator会在提交最终答案前，插入一步Verification
 - 2. Orchestrator在决定派发更强大的Subagent前，会思考：故障是执行问题还是子代理指令问题
 - 3. 如果是后者，则会重写子代理指令并重试，从而在不调用更昂贵模型的情况下得出正确答案。

Performance vs. Cost

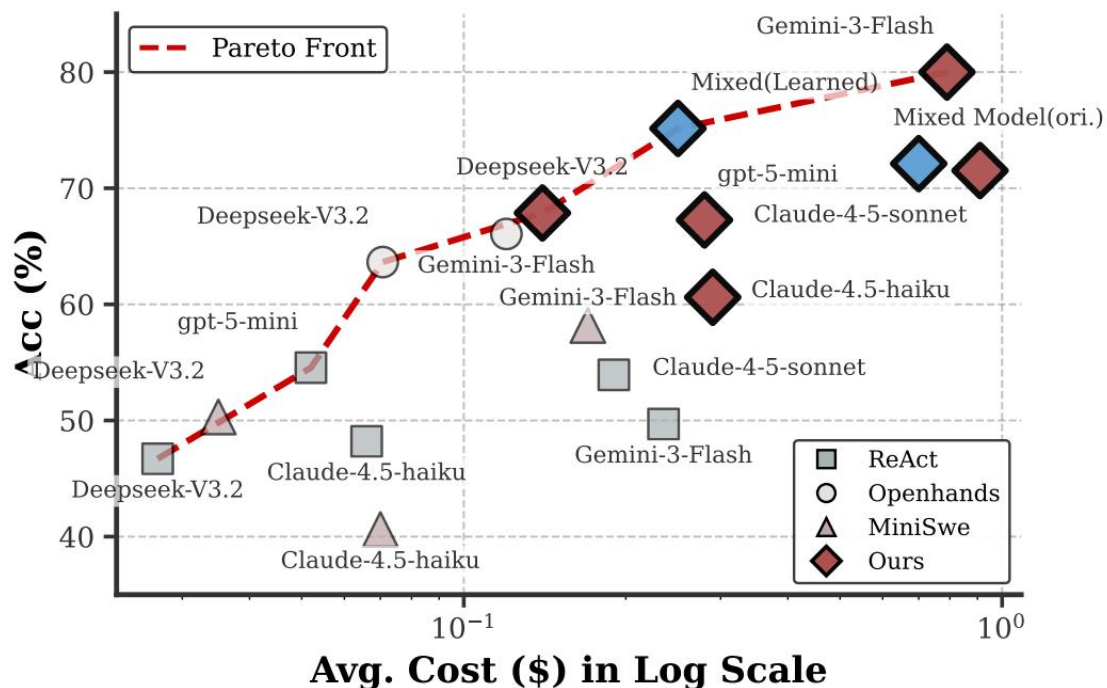


Figure 5. Pareto front curve of GAIA. We plot GAIA accuracy and average cost per task (USD, log scale). Each point corresponds to a configuration, and the dashed curve indicates the Pareto frontier formed by AORCHESTRA across different model routing choices.

- 帕累托曲线分析，本文方法可以达到帕累托最优：本文方法出现在曲线最外沿
- 展示不同的LLM, Agent Setting, 本文方法（是否使用Mixed LLMs, 是否采用ICL优化）

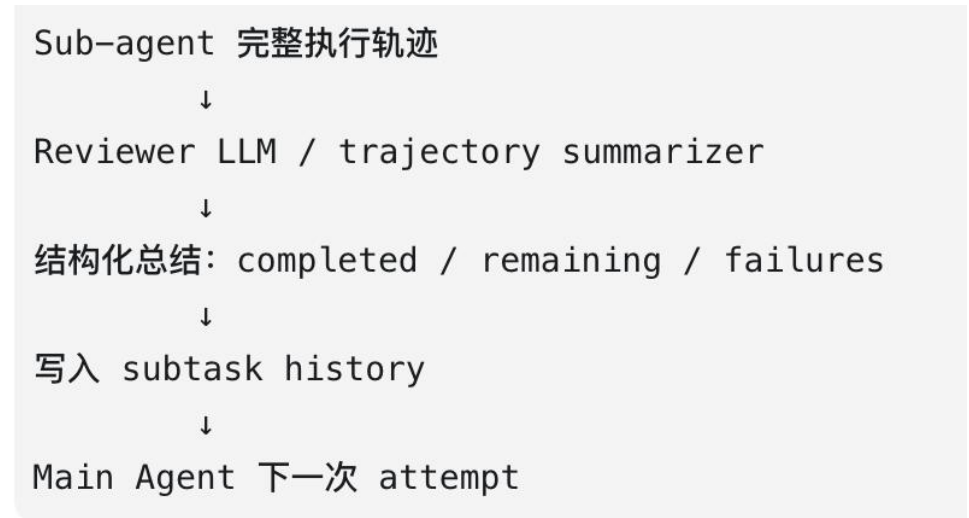
Experiments

Table 4. Evaluating plug-and-play sub-agents with a fixed orchestrator. We use Gemini-3-Flash to test the robustness and reusability of different sub-agent implementations.

System	Easy	Medium	Hard	Acc
Standalone baselines				
ReAct	50.00	34.09	16.67	28.57
Mini-SWE-Agent	50.00	40.91	20.83	34.29
Claude Code	50.00	41.86	16.67	32.86
Orchestrator with plug-in sub-agents				
ReAct-style SubAgent	50.00	63.63	20.83	48.57
Mini-SWE-style SubAgent	100.00	47.73	33.33	44.29

- Orchestrator的可复用性：替换不同风格的SubAgent Harness，仍获得一致的性能提升

- 开销过大，比如同gemini下，开销是ReAct的10倍
- 出于成本考虑，作者使用了子集而非全集进行评测
- SubAgent向Orchestrator交付结果时，需要一个reviewer LLM来总结轨迹、提取Insights等。而论文正文并未着重提及，也没有进行消融分析



Benchmark	全集	实际使用	覆盖率
GAIA validation	165	165	100%
Terminal-Bench 2.0	89	随机抽取70	78.7%
SWE-Bench Verified	500	随机抽取100	20%
Table 2 GAIA ablation	165	抽取50	30.3%